

B.Tech./ M.Tech. (Integrated) DEGREE EXAMINATION, MAY 2024
First & Second Semester

21LEH101T – COMMUNICATIVE ENGLISH
(For the candidates admitted from the academic year 2022-2023)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART - A (20 × 1 = 20Marks)Answer **ALL** Questions

- | | Marks | BL | CO | PO |
|---|-------|----|----|----|
| 1. Which of the following statement is true about communication?
(A) 50% of impact is determined by body language (B) 36% by voice tonality
(C) 7% by the words (D) 58% by kinesics | 1 | 2 | 1 | 10 |
| 2. If the marketing manager discusses about the present state of the work with the finance manager, it will be the case of _____.
(A) upward communication (B) downward communication
(C) horizontal communication (D) diagonal communication | 1 | 2 | 1 | 10 |
| 3. Which of the following is the process in which the receiver interprets and understands the message?
(A) Encoding (B) Decoding
(C) Channel (D) Source | 1 | 2 | 1 | 10 |
| 4. _____ listening is also known as critical listening.
(A) Active (B) Empathetic
(C) Appreciative (D) Evaluative | 1 | 2 | 1 | 10 |
| 5. Choose the correct verb forms for the following sentence:
The theatre company ____ a grant of £6.000 and it ____ a further £2.000 from elsewhere now.
(A) is given / has been seeking (B) will be given / seeks
(C) was given / has sought (D) has been given / is seeking | 1 | 3 | 2 | 10 |
| 6. Fill in the blanks with appropriate connectives:
We live in the same building; _____, we hardly see each other.
(A) Furthermore (B) However
(C) Therefore (D) Hence | 1 | 2 | 2 | 10 |
| 7. Fill in the blanks with correct form of verb that agrees with the subject:
A number of my friends _____riding bicycles, but neither my brother nor my sister _____a bicycle.
(A) love / owns (B) loves / own
(C) love / own (D) loves / owning | 1 | 2 | 2 | 10 |
| 8. Change the following sentence into passive voice:
I remember my sister taking me to the museum.
(A) I remember myself being taken to the museum by my sister. (B) I remember being taken to the museum by my sister.
(C) I remember taken to the museum by my sister. (D) I remember I was taken to the museum by my sister. | 1 | 2 | 2 | 10 |
| 9. Why is it important to use standardized symbols in flowcharts?
(A) To make the flowchart visually appealing (B) To represent artistic creativity
(C) To ensure clarity and understanding (D) To showcase individual style preferences | 1 | 3 | 3 | 10 |
| 10. Which of the following is an example of "flaming" in online communication?
(A) Complimenting someone's work in a group chat (B) Sending a friendly greeting to a new member of an online community
(C) Sharing helpful resources with colleagues (D) Engaging in a heated argument with insults and personal | 1 | 4 | 3 | 10 |
| 11. Which of the following is an example of synchronous communication?
(A) Email (B) Text messaging
(C) Forums (D) Video conferencing | 1 | 4 | 3 | 10 |
| 12. Choose the sentence with the correct question tag.
She went to the library yesterday, _____?
(A) didn't she (B) did she
(C) does she (D) is she | 1 | 4 | 3 | 10 |

13. What is the primary purpose of a letter of motivation in a job application? 1 3 4 10
 (A) To provide a summary of qualifications. (B) To request an interview.
 (C) To explain why you are the right candidate for the position. (D) To list previous work experiences.
14. In critical reasoning, what does it mean to "analyze the text"? 1 3 4 10
 (A) Summarizing the main points. (B) Criticizing the font style and size used.
 (C) Ignoring the content and focusing on grammar. (D) Identifying the author's argument and evidence.
15. **Find out the grammatical error in the following sentence:** 1 3 4 10
 The captain along with his team / are practicing very hard / for the / forth coming matches.
 (A) The captain along with his team (B) are practicing very hard
 (C) for the (D) forth coming matches.
16. **Identify the degree of comparison for the following sentence:** 1 3 4 10
 The roads of Chandigarh are much **wider** than the other cities of India.
 (A) Comparative (B) Positive
 (C) Superlative (D) Negative
17. Which of the following activities is an example of extensive reading? 1 3 5 10
 (A) Reading a scientific article to extract key findings. (B) Reading a novel for pleasure and general understanding.
 (C) Reading a legal document to understand specific clauses. (D) Reading a technical manual to learn specific procedures.
18. **Change the following direct speech into indirect speech:** 1 3 5 10
 The doctor said to the patient, "How do you feel now? Has the pain gone down?"
 (A) The doctor asked the patient how he had felt when the pain was gone. (B) The doctor asked the patient how he is feeling now and if the pain has gone down.
 (C) The doctor asked the patient how he felt then and if the pain had gone down. (D) The doctor asked the patient how he was feeling then and if the pain was down.
19. Which of the following is typically included in an agenda for a meeting? 1 3 5 10
 (A) Detailed meeting minutes (B) Background information on attendees.
 (C) Personal opinions of the meeting organizer. (D) Items to be discussed and their respective time slots.
20. An _____ report is a form of report in which reporters deal with a single topic of interest, such as serious crimes, political corruption, or corporate wrongdoing. 1 2 5 10
 (A) investigative (B) analytical
 (C) interrogative (D) Enquiry

PART - B (5 × 8 = 40 Marks)

Answer **ALL** Questions

21. **Read the following passage carefully and choose the best answer:**

Marks	BL	CO	PO
8	2	1	10

In a recent discussion paper, NITI Aayog has chalked out an ambitious strategy for India to become an artificial intelligence (AI) powerhouse. AI is the use of computers to make decisions that are normally made by humans. Many forms of AI surround Indians already, including chatbots on retail websites and programs that flag fraudulent bank activity. But NITI Aayog envisions AI solutions for India on a scale not seen anywhere in the world today, especially in five key sectors — agriculture, healthcare, education, smart cities and infrastructure, and transport. In agriculture, for example, machines will provide information to farmers on the quality of soil, when to sow, where to spray herbicide, and when to expect pest infestations. It's an idea with great potential: India has 30 million farmers with smart phones, but poor extension services. If computers help agricultural universities advise farmers on best practices, India could see a farming revolution. However, there are formidable obstacles. AI start-ups already offer some solutions, but the challenge lies in scaling these to cover the entire value chain, as NITI Aayog envisions. The first problem is data. Machine learning, the set of technologies used to create AI, is a data-guzzling monster. It takes reams of historical data as input, identifies the relationships among data elements, and makes predictions. More sophisticated forms of machine learning, like "deep learning", attempt to mimic the human brain. And even though they promise greater accuracy, they also need more data than what is required by traditional machine learning. Unfortunately, India has sparse data in sectors like agriculture, and this is already hampering AI-based businesses today. In fact, the lack of data means that deep learning doesn't work for all companies in India. One example is Climate-Connect, a Delhi-based firm, which uses AI to predict the amount of power a solar plant will generate every 15 minutes. This is critical because solar electricity generation can change dramatically every hour depending on weather conditions and the position of the sun. When this happens, the

plant must communicate expected changes to power distributors, which will then switch to alternative sources. With India planning to install 100 GW of solar power by 2022, such AI will play a central role in power planning. But to generate such data, Climate-Connect needs historical inputs like the time of sunrise and sunset, and cloud cover where the plant is located. Unfortunately, since most Indian solar plants are recent, data are available only for a couple of years, whereas deep learning needs data over many years to predict generation. Today, the firm uses traditional machine learning technologies such as regression analysis that work with less data. These methods have an accuracy of around 95%. While deep learning can boost accuracy for operations such as Climate-Connect, it hasn't worked very well in the Indian scenario, says Nitin Tanwar, cofounder of the firm. Another problem for AI firms today is finding the right people. NITI Aayog's report has bleak news: only about 50 Indian scientists carry out "serious research" and they are concentrated in elite institutions such as the Indian Institutes of Technology and the Indian Institutes of Science. Meanwhile, only about 4% of AI professionals have worked in emerging technologies like deep learning. A survey of LinkedIn found 386 out of the 22,000 people with PhDs in AI across the world to be Indians. How does this skill gap impact companies? To some extent, open libraries of machine learning code, which can be customised to solve Indian problems, help. This means that companies need not write code from scratch, and even computer science graduates can carry out the customisation.

- (i) What is the primary focus of NITI Aayog's ambitious strategy discussed in the passage?
 - a) Enhancing manual labour in agriculture.
 - b) Leveraging artificial intelligence across various sectors.
 - c) Improving rural infrastructure development.
 - d) Promoting traditional farming practices.
 - (ii) Which of the following sectors is NOT mentioned as a key focus area for AI solutions according to NITI Aayog?
 - a) Agriculture b) Healthcare c) Education d) Manufacturing
 - (iii) What is identified as a significant obstacle to scaling AI solutions in India?
 - a) Lack of government support
 - b) Insufficient technological infrastructure
 - c) Data scarcity in certain sectors
 - d) Overabundance of historical data
 - (iv) Which term is used to describe machine learning techniques that attempt to mimic the human brain?
 - a) Traditional learning b) Deep learning
 - c) Shallow learning d) Surface learning
 - (v) Which of the following is an antonym of the word **sparse**?
 - a) Scant b) Abundant c) Sporadic d) few
 - (vi) Which of the following weakens the argument for using more of AI powered tools in the future in India?
 - I. The AI sector uses a tremendous amount of electricity so as to process huge amounts of data which is not sustainable.
 - II. It is tough to collect, validate, standardize, correlate, archive and distribute AI-relevant data and make it accessible to organizations, people and systems.
 - III. Although AI will create more jobs than it would destroy.
 - a) Only I b) Only II c) Only I and II d) Only II and III
 - (vii) Which of the following statements weakens the argument about using 'Open Libraries' of machine learning code?
 - a) They contain material that can be used to solve issues.
 - b) Using such libraries is not a difficult job and does not need a higher level of understanding of coding.
 - c) It is possible to do a respectable amount of machine learning without mathematics.
 - d) These are not helpful in cases where there is neither a fixed algorithm nor a standard procedure.
8. What is cited as a critical factor for deep learning to predict solar power generation effectively?
- a) Availability of trained AI professionals
 - b) Government regulations on renewable energy
 - c) Historical data over many years
 - d) Technological advancements in solar panels

22. a. **Read the following Passage and draw the flow chart:**

The first stage in the recycling process for e-waste is the collection of electronic products through recycling bins, collection locations, take-back programs, or on-demand collection services. The mixed e-waste is then taken to

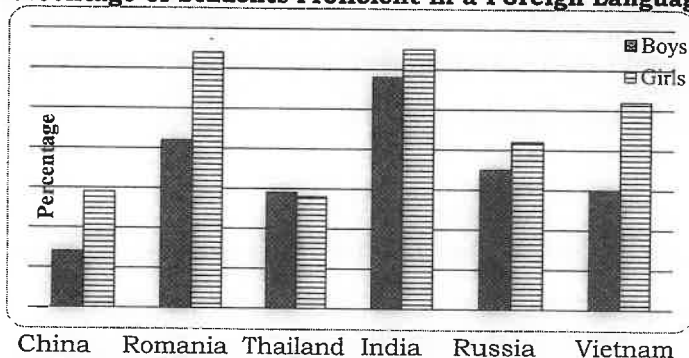
8 3 2 10

specialized electronics recyclers. Best practice dictates that e-waste should be separated by type at this stage of the process, which is why many collection sites will have different bins or boxes for different items. This is especially important for e-waste containing batteries, which require special treatment and can be very damaging if mixed with other waste. While safe storage may not appear critical, it can prove very important. For example, the glass screens of Cathode Ray Tubes (CRT) TVs and monitors are highly contaminated by lead. In the past, they were recycled into new computer monitors, but the growth of new technology and subsequent decline in demand for CRT products means much of this glass is now simply being stored indefinitely. E-waste then goes through the initial stage of manual sorting, where various items (such as batteries and bulbs) are removed for their own processing. This is the stage at which some items may also be manually dismantled for components, reuse, or the recovery of valuable materials. E-waste is then shredded into small pieces allowing for accurate sorting of materials, a key part of the process. Most electronics are a mix of materials, and breaking items down into pieces that measure just a few centimeters means they can be separated mechanically. The mechanical separation of the different materials actually consists of several processes one after the other. The two key steps are magnetic separation and water separation. The shredded e-waste is passed under a giant magnet, which is able to pull ferrous metals such as iron and steel from the mix of waste. In addition to this, an eddy current may also be used, separating the nonferrous metals. These materials can then be diverted to dedicated recycling plants for smelting. Other materials such as metal-embedded plastic and circuit boards are also separated at this stage. With a solid waste stream that now consists mainly of plastic and glass, water is used to separate the materials, further purifying for the separation of different plastics as well as hand-sorting obvious contaminants. The materials, now separated, are prepared for sale and reuse. For some materials, such as plastic or steel, this means joining another recycling stream. Others may be processed onsite and sold directly alongside usable components separated in the early stages.

(OR)

- b. Write a passage presenting the information contained in the bar graph using expressions of comparison and connectives of comparison and contrast. 8 3 2 10

Percentage of Students Proficient in a Foreign Language



23. a. **Write an essay of about 200 words on topics given below:** The Impact of Social Media on Mental Health. 8 4 3 10

(OR)

- b. **Write an essay of about 200 words on topics given below:** Netiquettes in the Digital Age.
24. a. Draft an email to the editor of 'The Hindu', Chennai, regarding the increase in the consumption of processed food and sedentary life style. 8 5 4 10

(OR)

- b. Write an extended definition for the following: 8 5 4 10
(i) Ethical Hacking (ii) Aerodynamics
25. a. Assume that you are the Director of Student Affairs of your College. Write a report on "The Decline of Reading Habits among the Students and its Impacts". 8 5 5 10

(OR)

- b. Elaborate on the essential skills for the effective-group discussion. 8 5 5 10

PART - C (1 × 15 = 15 Marks)

Answer **ANY ONE** Questions

- | | Marks | BL | CO | PO |
|---|-------|----|----|----|
| 26. Zuben Tech Pvt. Ltd. requires candidates with engineering background with a minimum of five years' experience. Detailed resume along with a cover letter should be sent to The HR Manager, Zuben Tech Pvt. Ltd., OMR Road, Chennai. | 15 | 6 | 4 | 10 |
| 27. Consider yourself as a secretary of the Robotics Club of your college. You conducted a meeting regarding the arrangements for the workshop. Draft a notice, agenda and minutes of the meeting. | 15 | 6 | 5 | 10 |
